

CSS 382 A: Introduction to Artificial Intelligence

Spring 2021 | Tuesday/Thursday 11:00AM-1:00PM

Instructor: Dr. Yusuf Pisan, pisan@uw.edu

Office Hours: Monday 1-2pm and Thursday 2-3pm. Signup via [Canvas Calendar](#)

Class Discord: [invite link](#)

Course Description

Principal ideas and developments in artificial intelligence, such as problem solving, knowledge representation, search, reasoning under uncertainty, learning, and natural language processing.

Learning Objectives

ABET Outcomes:

1. an ability to apply knowledge of mathematics, science, and engineering;
2. an ability to design a system, component, or process to meet desired needs within
3. realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability;
4. an ability to use the techniques, skills, and modern engineering tools;
5. knowledge of contemporary issues;

6. a recognition of the need for, and an ability to engage in lifelong learning;
7. an ability to identify, formulate, and solve engineering problems; and
8. an ability to design and conduct experiments, as well as to analyze and interpret data

At the end of this course, students will be able to:

1. Describe what artificial intelligence (AI) means and how machines can process information intelligently;
2. Identify the different fields that comprise AI as well as techniques and heuristics for each area;
3. Create a formal representation of a complex problem; and
4. Write programs to solve complex problems using AI methods in areas such as search, reasoning, natural language processing, games, and robotics.

Textbooks

[Optional Reference] Russell, Stuart J., and Norvig, Peter. Artificial Intelligence : a Modern Approach . Any edition, Pearson. <http://aima.cs.berkeley.edu/> and <https://www.amazon.com/Artificial-Intelligence-Approach-Stuart-Russell/dp/9332543518>

Grading

- Exercises: 25%
- Projects: 45%
- Midterm: 15%
- Final: 15%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% of the possible points will receive a 2.0 grade in this course. Students with 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

Distance Learning

Owing to the UW response to COVID-19, this course will be delivered entirely online through the Canvas learning management system and Zoom (supplemented by other web resources). This means that you will need to put in more time on your own than in a conventional course. Of course, the benefit of this structure is that you do not need to be physically present at UW-Bothell.

This course is scheduled to run synchronously at your scheduled class time via Zoom. These Zoom class sessions will be recorded. The recording will capture the presenter's audio, video and computer screen. Student audio and video will be

recorded if they share their computer audio and video during the recorded session. The recordings will only be accessible to students enrolled in the course to review materials. These recordings will not be shared with or accessible to the public. The University and Zoom have FERPA-compliant agreements in place to protect the security and privacy of UW Zoom accounts. Students who do not wish to be recorded should 1) change their Zoom screen name to hide any personal identifying information such as their name or UW netid, and 2) not share their computer audio or video during their Zoom sessions.

Policies

Pet Policy: If a pet enters the camera frame during class, we will pause our discussion for an introduction to that pet and admiration by all. All pets are welcome.

Assignment Submission: You can miss one exercise without penalty.

For projects, you can work individually or in pairs. When working as a pair

- Both students must enter a comment in the Canvas assignment with their name and their partner's name 72 hours before the assignment is due. Submit an empty file if necessary to leave a comment.
- All substantial work must be performed with both students present (virtually).

You can submit one and only one project 24 hours late without any penalty (if working as a pair both people will need to use their one-time extension). If you are using your 24 hour extension, you must post a brief explanation to "Using Extension" assignment before the assignment due date.

If you have any questions about your assignment grade, you must contact me within one week.

Plagiarism: Other than when working in pairs, talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the

student code of conduct:

<http://www.washington.edu/cssc/for-students/student-code-of-conduct/> I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

Any code or text that you copy from an external source must have a comment (or reference) indicating its origins.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. There will also be graded in-class group exercises to practice problem solving. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Communication: We will use discord as an extension of the classroom. Use a meaningful nickname and act professionally. If your question can be answered publicly by me or by a classmate, post it to discord. Use the office hours for complex issues or topics you are struggling with.

Use your UW email rather than "Canvas Messaging" to communicate directly with me. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Problems: If you are having difficulties, come and talk to me. If I don't know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Course Material: Lecture notes and other material will be posted to Canvas under "Files".

See the [School of STEM Course Policies](#), which covers:

- Academic Integrity
- Access and Accommodations
- Classroom Emergency Preparedness
- For Our Veterans

- Grade of Incomplete
- Inclement Weather
- Parenting Resources
- Religious Accommodations
- Respect for Diversity
- Student Support Services
- Surviving Sexual and Relationship Violence
- Wonder How to Address Faculty?

The UW Religious Accommodations Policy can be found at
<https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/>

Course Calendar

Topics and due dates below might be modified as needed.

See [Weekly Notes](#) for additional readings and pointers to external resources

Week	Tuesday/Thursday	Notes
	Week-1 Introduction, Uninformed Search	
	Week-2 A* Search and Heuristics	Search Project
	Week-3 Constraint Satisfaction	
	Week-4 Minimax, Expectimax, Utilities	Multi-Agent Search Project - Part1
	Week-5 Markov Decision Processes	Multi-Agent Search Project - Part2
	Week-6 Review Midterm	
	Week-7 Hidden Markov Models	Reinforcement Learning Project
	Week-8 Bayes' Nets	
	Week-9 Natural Language Processing	Ghostbusters Project
	Week-10 Perception, Robotics	Monday, 31 May, Memorial Day
	Final Final - Tuesday, 8 June	No classes during finals week